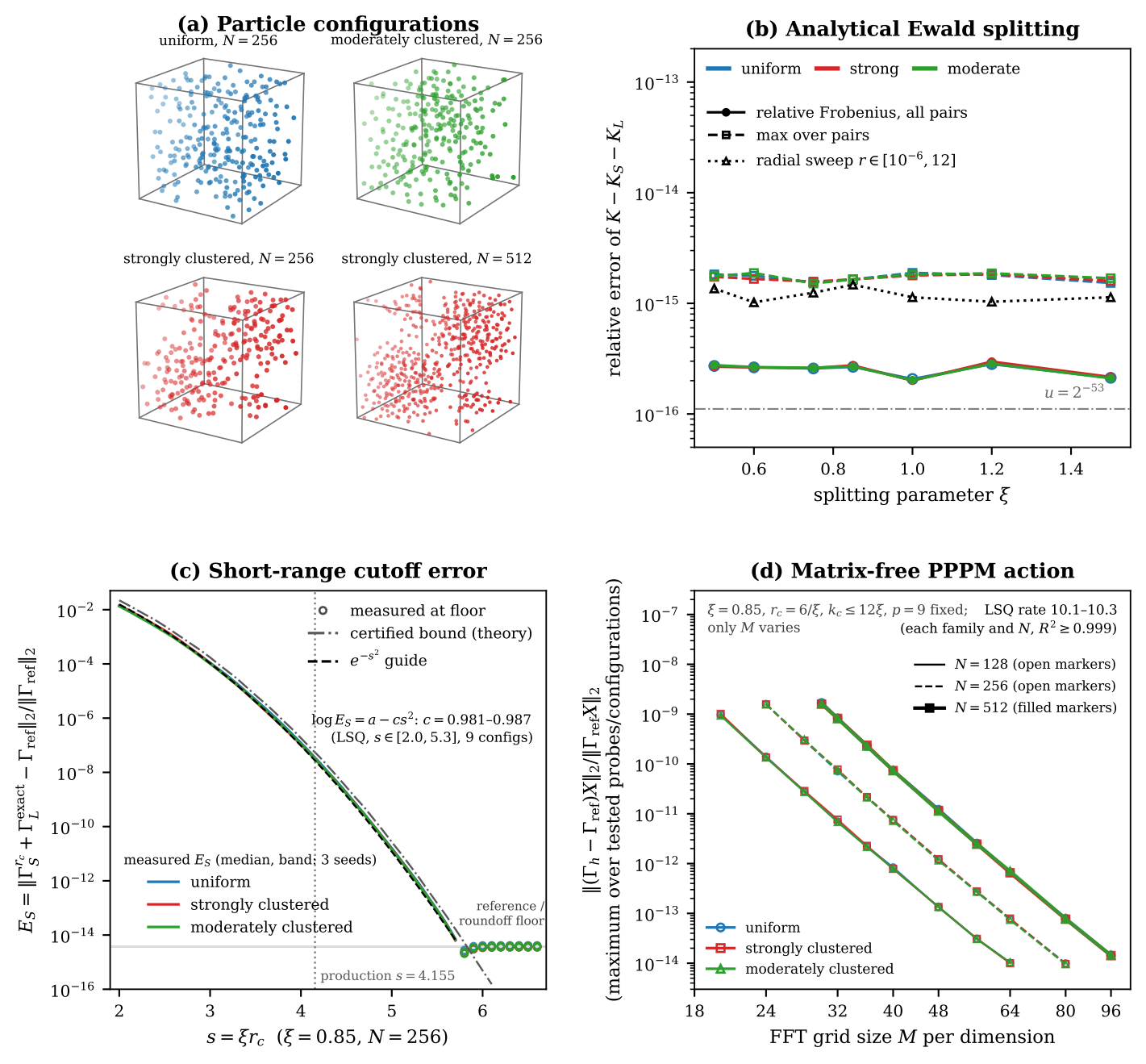

(a) Particle configurations
uniform, $N = 256$
moderately clustered, $N = 256$
strongly clustered, $N = 256$
strongly clustered, $N = 512$
(b) Analytical Ewald splitting
uniform
strong
moderate
relative Frobenius, all pairs
max over pairs
radial sweep $r \in [10^{-6}, 12]$
$u = 2^{-53}$
relative error of $K - K_S - K_L$
splitting parameter ξ
(c) Short-range cutoff error
measured at floor
certified bound (theory)
e^{-s^2} guide
$\log E_S = a - cs^2$: $c = 0.981$–0.987
(LSQ, $s \in [2.0, 5.3]$, 9 configs)
measured E_S (median, band: 3 seeds)
uniform
strongly clustered
moderately clustered
reference / roundoff floor
production $s = 4.155$
$E_S = \|\Gamma_S^{r_c} + \Gamma_L^{\text{exact}} - \Gamma_{\text{ref}}\|_2 / \|\Gamma_{\text{ref}}\|_2$
$s = \xi r_c$ ($\xi = 0.85$, $N = 256$)
(d) Matrix-free PPPM action
$\xi = 0.85$, $r_c = 6/\xi$, $k_c \le 12\xi$, $p = 9$ fixed; only M varies
LSQ rate 10.1–10.3
(each family and N, $R^2 \ge 0.999$)
$N = 128$ (open markers)
$N = 256$ (open markers)
$N = 512$ (filled markers)
uniform
strongly clustered
moderately clustered
$\|(\Gamma_h - \Gamma_{\text{ref}})X\|_2 / \|\Gamma_{\text{ref}}X\|_2$
(maximum over tested probes/configurations)
FFT grid size M per dimension